PSI Center for Scientific Computing, Theory and Data

CFD–GEMS3K modelling of iodine transport and deposition

LESTO Task 3.4 progress

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From column experiments to the reactor hall

Task 3.4: fission products from the source into the reactor hall

Why thermal-gradient columns

- volatile iodine compounds released from LBE travel with a gas and deposit where they become supersaturated
- columns do this under control: known gas flow, measured wall temperature, measured deposit along the tube
- a model that reproduces them has its transport, wall exchange and chemistry tested together

Where this tool fits in the task

- PSI: CFD (OpenFOAM) coupled with GEMS chemical equilibrium; experiments; later machine learning
- SCK CEN: a comparative approach, HSC Chemistry with Fluent
- CIRTEN: reduced-order models
- VKI is also a partner of the task

This talk: the PSI CFD–GEMS3K model and what it can offer to the other approaches.

Two sets of published column experiments

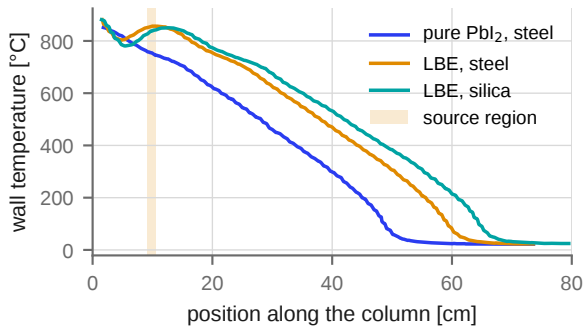
Eight experiments with measured wall temperature and deposit scans

Liu et al. (2025), PSI

- 1.25 m columns: steel (4.8 mm) and silica (5.0 mm)
- pure PbI_2; iodine in LBE at 900 °C
- helium 45 or 100 mL/min, 3 h; 1 cm scans

Lobresco et al. (2026)

- five pure PbI_2 experiments: 5–540 mL/min, 6–300 s

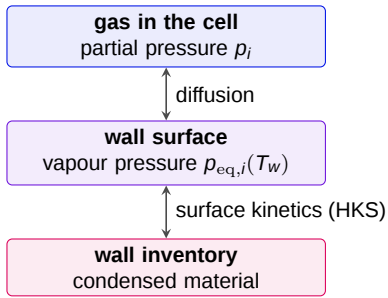


Digitised from Liu et al. (2025)

Model, part 1: transport and wall exchange

Trace gases in laminar helium; a conservative law for the wall

- OpenFOAM finite volumes, axisymmetric 5° wedge of the column
- laminar helium carrier ($\mathrm{Re} \approx 0.7$–3), computed once and frozen
- vapours are dilute traces carried and diffusing in it
- wall exchange: Hertz–Knudsen–Schrage evaporation/condensation law in series with diffusion to the wall
- condensed material is stored in wall inventories that can evaporate again

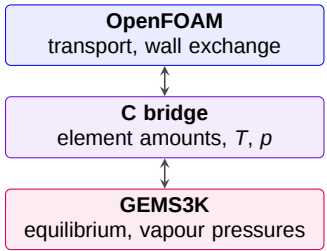


Two resistances in series: whichever is larger controls the uptake.

Model, part 2: chemistry with GEMS3K

Nine Pb–Bi–I gases re-equilibrate in every cell; no atom is lost

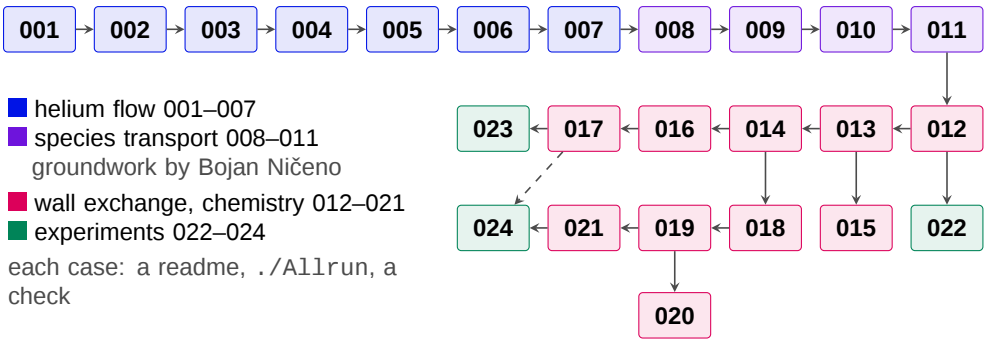
- gases: Pb, PbI, PbI_2, Bi, Bi_2, BiI, BiI_3, I, I_2
- condensed: PbI_2, BiI_3, Pb, Bi on the wall
- each step: transport, then local gas equilibrium at fixed element amounts of Pb, Bi and I
- GEMS3K (PSI's Gibbs-energy minimisation code) linked into the solver through a small C bridge
- an element ledger checks every atom



Function calls in memory, no file exchange per step.

Built step by step: 24 test cases

Every capability has a case that isolates it and a number that checks it



- helium flow 001–007
- species transport 008–011
  groundwork by Bojan Ničeno
- wall exchange, chemistry 012–021
- experiments 022–024

each case: a readme, `./Allrun`, a check

Verified: wall law and conservation

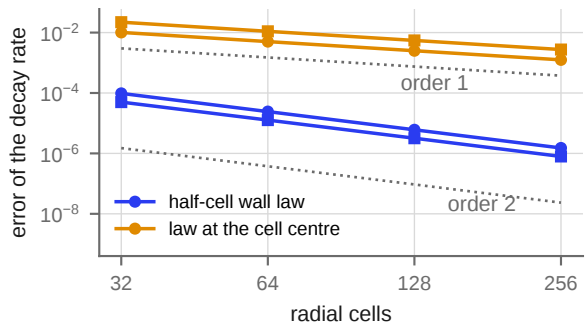
The wall law is second-order accurate; element balances close at round-off

Wall law against an exact solution

- pipe flow with wall uptake (Graetz–Robin), exact decay rate
- half-cell wall law: observed order 1.99–2.00
- law at the cell centre: order 1.004–1.008, 100× larger errors

Conservation

- worst element residue: 1.41×10^{-12} of the supplied material (19 production runs)



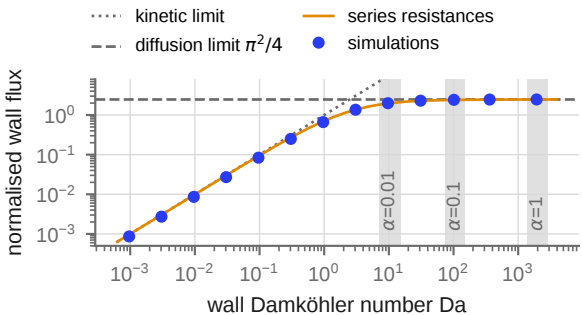
45 runs; exact reference

Deposition is limited by diffusion

Both limits appear; the columns sit on the diffusion side

- Da: surface conductance over diffusion to the wall
- small Da: kinetics decide; large Da: diffusion limit $\pi^2/4$
- onset stays at 561 K: saturation sets it
- columns: Da $\approx$ 1.4–2.9$\times10^3$ (α = 1), 76–150 (0.1), 7–15 (0.01)

So: the sticking coefficient hardly matters; diffusivity, wall temperature and vapour pressure do.



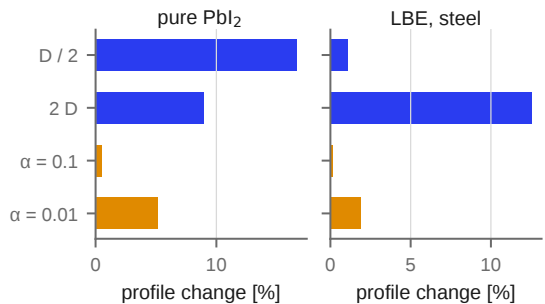
Test channel, 13 runs; grey: the column cases

What matters for the deposit

Diffusivity, temperatures and thermodynamic data matter; kinetics does not

Change of the iodine wall profile (L_1, %):

diffusivity halved or doubled	up to 16.65
flow reference 273 vs 298 K	up to 10.93
silica temperature curve shifted 4.1–4.65 cm	187–194
BiI(g) data: BiI_3 share of iodine	94.6 → 2.0 %
accommodation 0.1 or 0.01	at most 5.14
source moved ±1.5 cm	at most 0.19



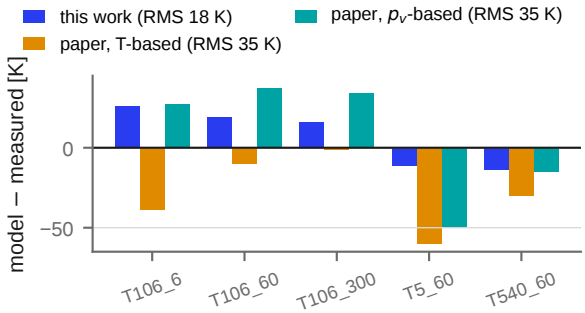
Two cases repeated with the production scheme

Lobresco et al.: onset within 18 K

Five PbI_2 experiments, untuned: onset RMS 18 K; both published models 35 K

- no parameter fitted to these data
- onset: 1 % of the deposit peak, as in the paper
- computed peaks lie 1–4 cm upstream
- carrier, vapour pressure and source differ from the published models

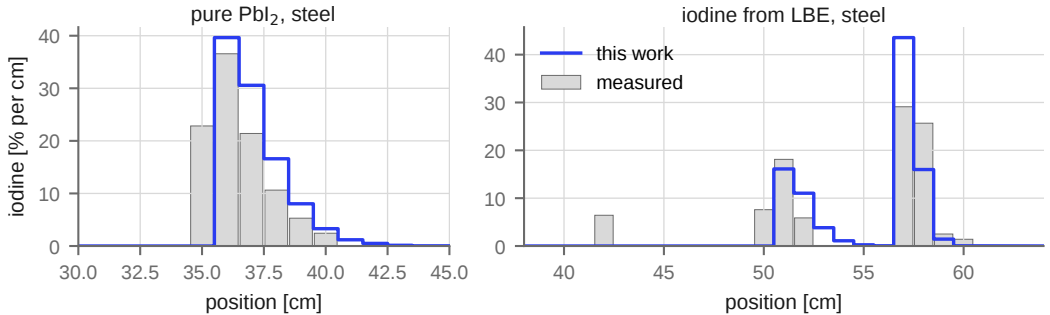
Promising, conditional: not yet attributable to the wall law alone.



Data and models: Lobresco et al. (2026), CC BY 4.0

Liu et al.: the steel columns

PbI_2 peak 366.5 °C (measured 368 ± 18 °C); both LBE bands

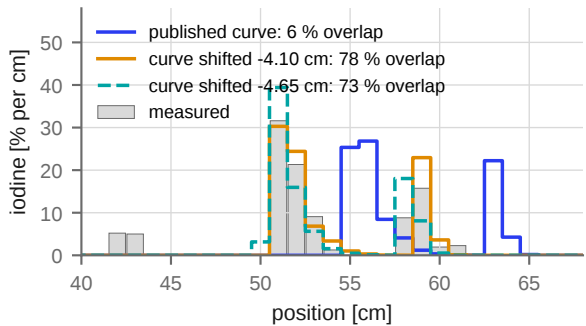


Measured scan adapted from Liu et al. (2025). Right: a PbI_2 band near 51 cm, a BiI_3 band near 57 cm.

Liu et al.: the silica column

The printed peak temperatures do not match the printed temperature curve

- published curve: computed bands 4–5 cm downstream
- printed peak labels (306, 127 °C) sit 4.1–4.65 cm off the curve
- curve shifted by that: overlap 6 % → 73–78 %
- a hypothesis, not a result: the thermocouple data would settle it

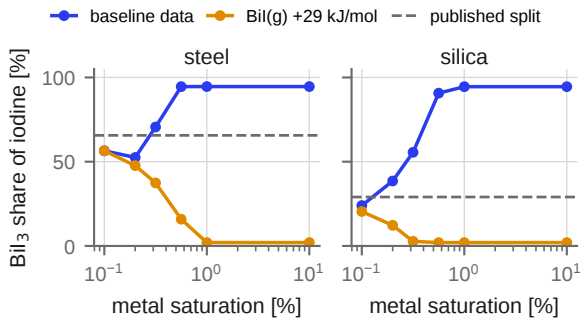


Published data adapted from Liu et al. (2025)

Thermodynamic data decide the iodide

One BiI(g) value moves the BiI_3 share of iodine from 94.6 to 2.0 %

- baseline data vs BiI(g) shifted by +29 kJ/mol
- steel, 1 % metal saturation: 94.58 vs 2.01 %
- the same at every tested resolution
- not an uncertainty band: qualified Bi–I data are needed



LBE with added metal vapour; published split: Liu et al. (2025)

D38: the coupled tool and what it offers

GEMS3K in every cell at production scale; cases to share

Status

- D38 (M24): the coupled GEMS and CFD codes, description of the coupling and modelling; draft ready
- production case with GEMS3K data in every cell: profiles change by at most 2.1×10^{-4}
- 5.1×10^{8} GEMS3K calls, none failed

What we can offer

- reference cases with measured inputs for comparative CFD, e.g. with HSC Chemistry and Fluent
- reference and training data for reduced-order models
- a verified wall-exchange law and a conservative coupling, documented case by case

What the model does not yet cover

A conservative, verified method; not yet independent predictive validation

Inputs

- release history, flow reference conditions and coordinate origins are assumed
- near the source the vapour mole fraction reaches 0.33–0.36: not dilute there
- boat and end of the experiment (cooling, withdrawal) not simulated

Physics and numerics

- frozen carrier: no buoyancy; Richardson number up to about 1.2 in the steepest section
- no aerosols, no thermophoresis
- mesh: one case refined with the production scheme (1.98 %); with upwind four cases change by 2.13–7.30 %

Where partners could help

Data and comparisons on common cases would turn this into a validation

Data

- measured wall-temperature tables and scan coordinates
- release histories or boat mass before and after
- repeat measurements: experimental scatter
- series at different flow rates or durations
- single-compound runs with a known released amount

Thermodynamics and models

- qualified Bi–I records, in particular BiI(g), and their redistribution terms
- comparative calculations on the same cases with other codes
- reduced-order models tested against the CFD runs

Next steps

Outputs

- methods paper on the CFD model: submission to Physics of Fluids planned
- deliverable D38 (M24): the coupled GEMS and CFD codes
- the case repository: every case reruns with one command

Work ahead

- further validation cases, with measured inputs
- common reference cases with the partners

Thank you

Questions, data and ideas are welcome.

Jan Kren, PSI, LESTO Task 3.4

PSI

Backup

Backup: spatial and iteration accuracy

Mesh and advection scheme

- production: bounded limited-linear advection, $576 \times 72 = 41{,}472$ cells
- first-order upwind differs by 7.7–20.8 % at equal mesh (steel case of the scheme study)
- 41,472 $\rightarrow$ 93,312 cells: 1.98 % (bounded, one case); 2.13–7.30 % (upwind, four cases)
- apparent orders inconsistent: no formal grid-convergence index

Time step and iterations

- time step halved: iodine profiles change by at most 0.323 %
- eight instead of three species correctors over 60 s: at most 0.021 % (four cases)
- 41,472 $\rightarrow$ 93,312 cells (upwind): total deposits change by less than 0.02 %

Backup: omitted effects, bounded

Buoyancy

- Grashof number ≤ 2.2; Reynolds number of order one
- Richardson number 0.15–0.78 between 400 and 750 K, about 1.2 in the steepest cold section
- compositional Richardson number ≤ 0.32
- column orientation not published: an unquantified limitation

Thermal (Soret) diffusion

- thermal diffusion factor 1 as a generous bound
- axial drift $\leq 2.2\,\%$ of the carrier velocity
- radial flux $\leq 0.7\,\%$ of the Fickian flux
- wall-to-bulk temperature difference ≤ 2.6 K